

Research Assistant/Postdoctoral Research Assistant

About the Role

The School of Electronic Engineering and Computer Science at Queen Mary University of London is looking to recruit a Postdoctoral Research Assistant to support the projects led by **Dr. Ahmed Sayed**. The main project (KUber) is, EPSRC-funded, aiming at building novel distributed architectures to facilitate the exchange and delivery of acquired knowledge among machine learning entities at scale. For brief details of projects including KUber, plz see <https://kuber.org.uk>

About You

Applicants should preferably have an MSc/PhD in Computer Science/Engineering. This is mainly aimed at PhD graduates or PhD students nearing their PhD completion. They should have expertise in distributed AI/ML systems and computer networks. Good knowledge and practical skills in distributed AI/ML, networked systems, data management, and/or software engineering are recommended for this position. The post requires strong research, programming and system development abilities. Prior involvement in notable open-source projects is a plus. Applicants should have publications in leading AI/ML Systems conferences and journals in the fields above.

Benefits

The post is based at the Mile End Campus in London. It is a full-time, fixed-term appointment until 31-Mar-2027. The starting salary will be Grade 4, depending on qualifications and experience, including London Allowance.

Informal enquiries should be addressed to **Dr. Ahmed Sayed** (www.eecs.qmul.ac.uk/~ahmed) at ahmed.sayed@qmul.ac.uk

Formal Applications will open soon

Some Relevant Recent Publications

Conferences:

Norah Alballa, Wenxuan Zhang, Ziquan Liu, **A. M. Abdelmoniem**, Mohamed Elhoseiny, Marco Canini, Query-based Knowledge Transfer for Heterogeneous Learning Environments, **ICLR** 2025
Ahmad Faraz Khan, Samuel Fountain, **A. M. Abdelmoniem**, Ali R Butt, Ali Anwar, FLStore: Efficient Federated Learning Storage for non-training workloads, **MLSys** 2025
Ahmad F Khan, Azal A Khan, **A. M. Abdelmoniem**, Samuel Fountain, Ali R Butt, Ali Anwar, FLOAT: Federated learning Optimisations with automated tuning, **ACM EuroSys** 2024
Q Li, **A. M. Abdelmoniem**, Mitigating Malicious Attacks in Federated Learning via Confidence-aware Defense, arXiv preprint arXiv:2408.02813, 2024
A. M. Abdelmoniem, AN Sahu, M Canini, SA Fahmy. "REFL: Resource Efficient Federated Learning". In Proceedings of ACM **EuroSys**, Rome, Italy, 2023

Journals:

A. M. Abdelmoniem and Y. Zhao, "Knowledge Routing in Decentralized Learning," in **IEEE Intelligent Systems**, 2025
A. M. Abdelmoniem, Mona Jaber, Ali Anwar, Yuchao Zhang, Mingliang Gao, Towards a Decentralized Collaborative Framework for Scalable Edge AI, **Future Internet**, 2025
Wentai Wu, Ligang He, Saiqin Long, **Ahmed M. Abdelmoniem**, Yingliang Wu, Rui Mao, Knowledge Augmentation in Federation: Rethinking What Collaborative Learning Can Bring Back to Decentralized Data, ArXiv 2503.03140, 2025
Amna Arouj, **Ahmed M. Abdelmoniem**, Towards Energy-Aware Federated Learning via Collaborative Computing Approach, **Computer Communications**, Volume 221, 2024.
Z. Tian, Z. Wang, **A. M. Abdelmoniem**, G. Liu and C. Wang, "Knowledge Representation of Training Data With Adversarial Examples Supporting Decision Boundary", **IEEE Transactions on Information Forensics and Security (IEEE TIFS)**, 2023.
A. M. Abdelmoniem, Chen-yu Ho, Pantelis Papageorgiou, Marco Canini. "A Comprehensive Empirical Study of Heterogeneity in Federated Learning". **IEEE Internet of Things Journal**, 2023.